**now three main things it need to take care first it needs to take care of the attendance thing for this the structure it is human feed back in the loop it will fist take the time table since it is asking for time table create a new feild for every semester now what you will do is ask the user for the feedback of every users classes that are held every week by taking a human feedback in the loop also taking care of holidays or any teacher did not attended the class also accordingly create a tracker of the attendance like in most colleges it is >= 75% fasking it from the user and give a warning accordingly to the user based on trigger system between range absolute(current_attendance_mainted - 75%)<=10% then the next thing setting up a simple exam reminder before each exam in the evening and this will be done only when the user uploads any exam schedule also verify that he has uploaded the latest exam schedule not any outdated one and then all these notifications can be send via a discord bot on the users personal channel or to the user itself if possible also make sure that it also keeps track of the assignment along with the deadline and send reminder in the evening before the deadline and mark it as missed via google classroom if not detected as submitted if any that the user recieves on the google classroom also make sure to manage normal ai suggestions related to student's routine, summarize important**

**updates, organize schedules intelligently, answer campus-related questions**

**instantly, and proactively help students manage everyday academic and**

**personal life?**

### 1. Brainstorming & Setting Goals

**Goal:** Build an AI-powered student assistant that prevents students from missing classes, exams, assignments, and important campus updates.

Core problem:
Students receive academic information from too many scattered places: timetable PDFs, Google Classroom, WhatsApp groups, college portals, exam schedules, and verbal updates. Because of this, they miss deadlines, lose attendance, and feel stressed.

Main goals:

- Track attendance accurately using timetable + human feedback.
- Warn students when attendance approaches the 75% danger zone.
- Remind students about exams and assignments at the right time.
- Summarize important academic updates.
- Send daily actionable notifications through Discord.
- Keep humans in the loop so wrong timetable/exam/class data does not break the system.

Success metrics:

- Reduce missed assignments.
- Reduce attendance shortage surprises.
- Reduce time spent checking multiple platforms.
- Improve student routine planning.

### 2. Requirements Analysis & Plan

**Functional requirements:**

- Student can create semester profile.
- Student can upload timetable.
- Student can upload exam schedule.
- System extracts subjects, class timings, exam dates, and deadlines.
- System asks weekly/daily feedback: class held, cancelled, holiday, teacher absent, attended/not attended.
- System calculates attendance per subject.
- System triggers alerts when attendance is near or below 75%.
- System connects with Google Classroom to track assignments.
- System sends Discord reminders.
- AI summarizes important updates and answers campus-related questions.

**Non-functional requirements:**

- Low-cost AWS hosting.
- Privacy-first student data handling.
- Human confirmation for extracted schedules.
- Fast response time for reminders and Q&A.
- Reliable evening notification system.
- Explainable alerts: every alert should show why it was triggered.

**Main users:**

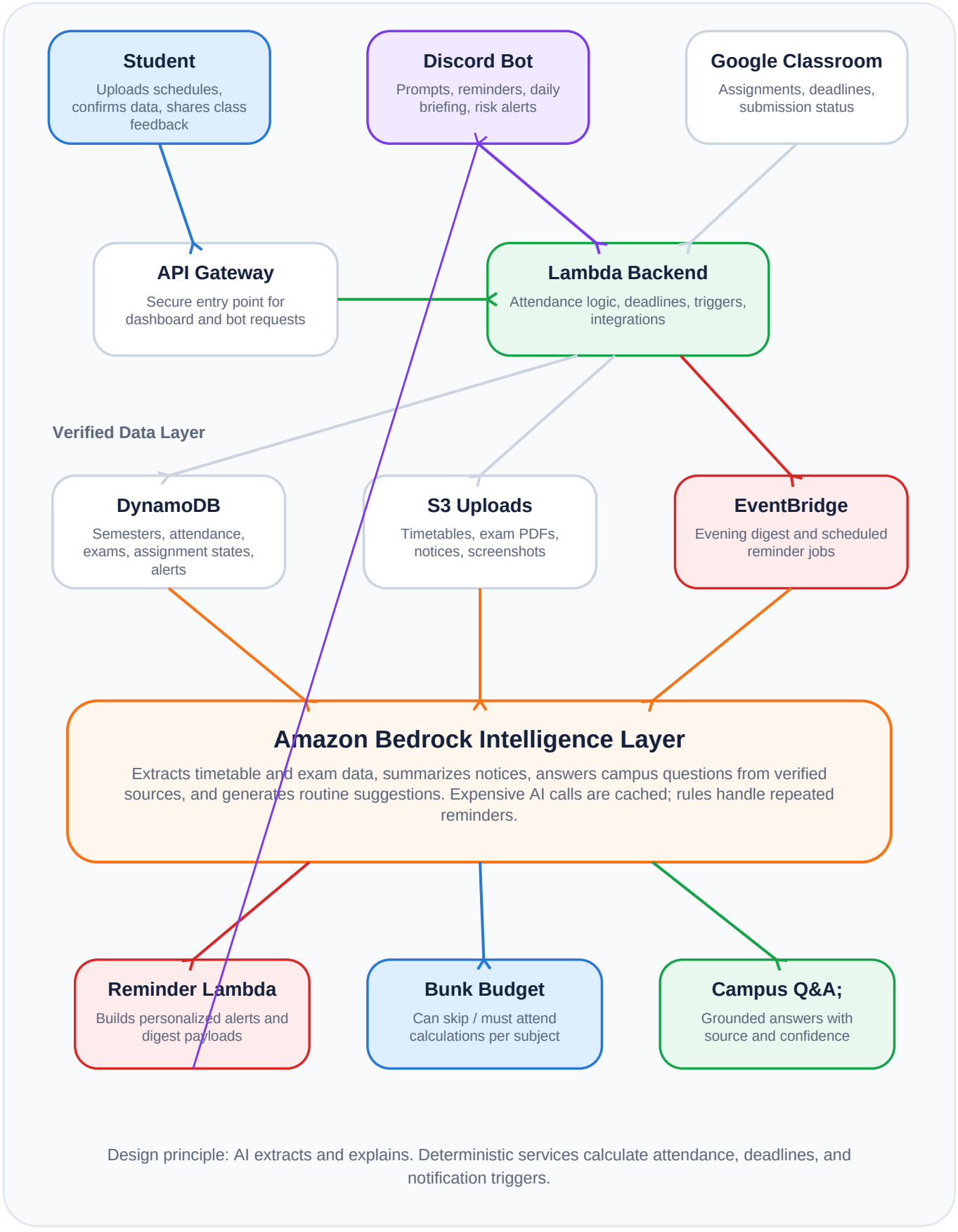
- Student
- Class representative / group admin, optional later
- Faculty/admin, future expansion

**MVP scope:**

- Semester setup
- Timetable upload
- Attendance tracker
- Exam reminders
- Assignment reminders
- Discord daily briefing
- Basic AI campus Q&A

# CampusFlow Architecture

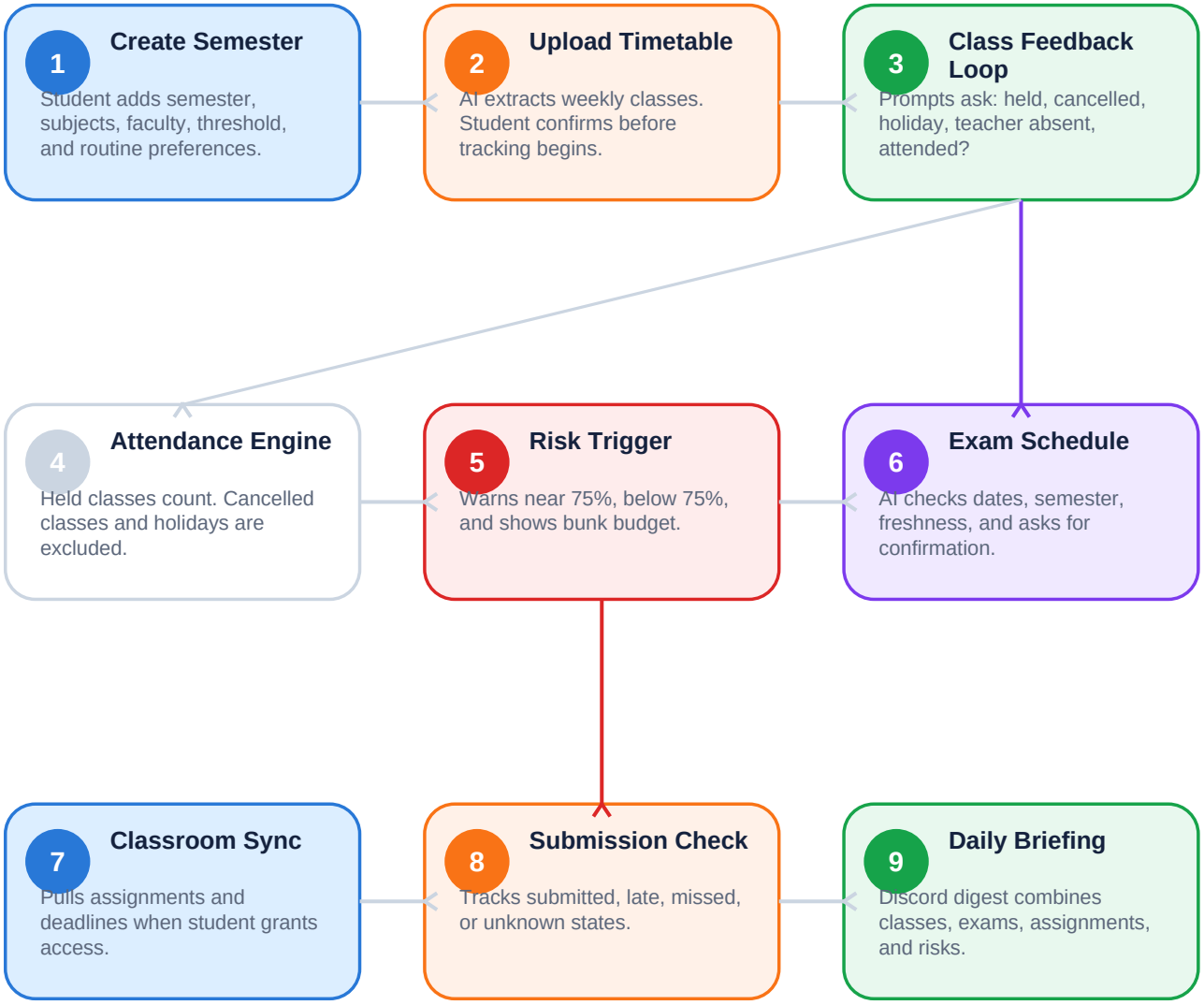
Serverless AWS design with verified student data, low-cost automation, and AI only where it adds judgment




Design principle: AI extracts and explains. Deterministic services calculate attendance, deadlines, and notification triggers.

# CampusFlow Workflow

From messy college inputs to verified attendance, deadline reminders, and daily student guidance




## Human-in-the-loop safety

Every extracted item stores source, confidence, and last verified date. This prevents outdated exam schedules, false class counts, and unsupported campus answers from reaching the student as facts.